

JOB ID: 523676

Integration Test Software Engineer

Position Location: This role supports our Parsippany-Troy Hills, NJ branch location, which serves customers throughout the greater Parsippany-Troy Hills, NJ area.

Here at Siemens, we take pride in enabling sustainable progress through technology. We do this through empowering customers by combining the real and digital worlds. We are improving how we live, work, and move today and for the next generation! We know that the only way a business thrives is if our people are thriving. That's why we always put our people first. Our global, diverse team would be happy to support you and challenge you to grow in new ways. Who knows where our shared journey will take you?

Our Smart Buildings help to create efficient, safe, adaptable, and responsible environments. Our aim isn't just about improving buildings; it's about creating perfect places that improve people's lives.

Transform the everyday with us!

We are looking for a hands-on **Integration Test Software Engineer** for our Parsippany, NJ office who enjoys software development, automation, embedded platforms, and practical problem solving. In this role, you will validate integrated software running on Siemens fire safety systems from component integration through release readiness, drive automated test framework adoption, and apply AI methods where they can responsibly improve coding, test creation, debugging, analysis, and decision support. You will gain deep exposure to safety-critical embedded product development while helping deliver dependable solutions to the market.

As an Integration Test Software Engineer, you will:

- Validate integrated software on target fire safety hardware, including embedded firmware, application software, communication interfaces, and connected services
- Improve requirements testability, traceability, defect resolution, and release readiness for products whose reliability can directly affect people's safety
- Advance automated integration test frameworks that improve speed, repeatability, coverage, scalability, and release confidence
- Apply AI-assisted engineering methods, including tools such as GitHub Copilot and emerging agentic systems, to improve coding, test generation, troubleshooting, and analysis in critical fire safety applications where reliability, evidence, and human oversight are essential
- Design, develop, execute, and maintain integration test plans, test cases, automated test scripts, and supporting software tools for fire safety products.
- Drive automated test framework development, including reusable libraries, test harnesses, simulation interfaces, data collection, reporting, and CI/CD integration.

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- Use Python or similar scripting languages to build tools for test automation, data analysis, diagnostics, and workflow improvement.
- Apply AI-assisted development and testing methods, including GitHub Copilot, development copilots, and agentic systems, to support code development, test generation, log analysis, failure-pattern recognition, defect triage, and coverage assessment.
- Evaluate emerging AI tools with appropriate validation, documentation, cybersecurity awareness, and human oversight before adoption in safety-critical workflows.
- Verify integrated software behavior on target hardware across embedded firmware, application software, communication interfaces, connected services, and networked devices.
- Perform functional, regression, interoperability, end-to-end, stress, fault-injection, scalability, and reliability testing.
- Validate software integration with fire alarm control panels, detectors, notification appliances, gateways, networked devices, and related software services on representative target hardware.
- Apply knowledge of communication protocols and networks such as CAN, Ethernet/IP, TCP/IP, RS-485, HNET/XNET, and FDnet.
- Troubleshoot software integration defects, communication failures, automation failures, and target-hardware issues; support defect triage, root cause analysis, release qualification, and field validation as needed.
- Use EWM/ETM or similar test management tools to plan, execute, document, and track testing activities.
- Review requirements to improve testability and maintain traceability between requirements, test cases, defects, automation evidence, and release documentation.
- Demonstrate:
 - Strong software engineering mindset, solid programming capability, and curiosity for automation, simulation, software architecture, and AI-assisted development.
 - Analytical troubleshooting across embedded software, target hardware, communication networks, and test infrastructure.
 - Commitment to continuous learning as AI, automation, embedded systems, and connected fire safety technologies evolve.
 - Clear communication and collaboration across software, firmware, hardware, systems, quality, and product teams.
 - Commitment to quality, reliability, and the responsibility that comes with life-safety systems.

You will make an impact with these qualifications:

Basic Qualifications:

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- Bachelor's degree in electrical engineering, computer engineering, software engineering, computer science, or a related technical field.
- 2+ years of experience in embedded software, software integration testing, target-hardware validation, test automation, or software development for technical products.
- Programming skills in Python and at least one additional language commonly used in embedded or test automation environments, such as C, C++, C#, Java, or JavaScript/TypeScript.
- Must be able to demonstrate:
 - Understanding of software development practices and software architecture concepts, including modular design, interfaces, APIs, version control, build pipelines, and maintainable testable code.
 - Ability to read requirements, define test scenarios, analyze test results, and clearly document defects and evidence.
 - Strong troubleshooting mindset and interest in learning complex embedded software, target hardware, communication networks, automated testing, and AI-enabled engineering workflows.
- Qualified applicants must be legally authorized for employment in the United States on a continual and permanent basis without company sponsorship.

Preferred Qualifications:

- Knowledge of embedded fire safety platforms such as Cerberus PRO, FS20M, or similar target-hardware-based systems.
- Familiarity with compliance and regulatory testing, including NFPA 72, UL 864, UL 521, or UL 268.
- Understanding of embedded software and hardware platforms, including microcontrollers, firmware, software upgrades, target hardware, and communication interfaces.
- Experience building, maintaining, or improving automated test environments, execution workflows, and CI/CD-connected test infrastructure.
- Experience with AI-assisted coding tools such as GitHub Copilot or similar development copilots.
- Exposure to agentic AI systems, automated code or test generation, AI-assisted debugging, or AI-supported engineering workflows.
- Understanding of the risks, validation needs, cybersecurity considerations, and human oversight required when applying AI methods in regulated or safety-critical products.
- Exposure to secure communications, vulnerability assessment, or security-minded product validation.
- Comfort managing multiple priorities in an Agile product development environment.

#LI-TAG #HQ-EREF #VeteranCareers #TransitioningServiceMember #MilitarySpouse

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Ready to create your own journey? Join us today!

About Siemens

We are a global technology company focused on industry, infrastructure, transport, and healthcare. From more resource-efficient factories, resilient supply chains, and smarter buildings and grids, to sustainable transportation as well as advanced healthcare, we create technology with purpose adding real value for customers.

Our Commitment to Equity and Inclusion in our Diverse Global Workforce

We value your unique identity and perspective. We are fully committed to providing equitable opportunities and building a workplace that reflects the diversity of society, while ensuring that we attract the best talent based on qualifications, skills, and experiences. We welcome you to bring your authentic self and transform the everyday with us.

You'll Benefit From

Siemens offers a variety of health and wellness benefits to our employees. Details regarding our benefits can be found here: <https://www.benefitsquickstart.com/siemens/index.html>

The pay range for this position is \$67,616 - \$115,913 annually. The actual wage offered may be lower or higher depending on budget and candidate experience, knowledge, skills, qualifications, and premium geographic location.

Equal Employment Opportunity Statement

Siemens is an Equal Opportunity Employer encouraging inclusion in the workplace. All qualified applicants will receive consideration for employment without regard to their race, color, creed, religion, national origin, citizenship status, ancestry, sex, age, physical or mental disability unrelated to ability, marital status, family responsibilities, pregnancy, genetic information, sexual orientation, gender expression, gender identity, transgender, sex stereotyping, order of protection status, protected veteran or military status, or an unfavorable discharge from military service, and other categories protected by federal, state or local law.

EEO is the Law

Applicants and employees are protected from discrimination on the basis of race, color, religion, sex, national origin, or any characteristic protected by Federal or other applicable law.

Reasonable Accommodations

If you require a reasonable accommodation in completing a job application, interviewing, completing any pre-employment testing, or otherwise participating in the employee selection process, please fill out the accommodations form by [clicking on this link](#). If you're unable to complete the form, you can reach

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out to our AskHR team for support at 1-866-743-6367. Please note our AskHR representatives do not have visibility of application or interview status.

Pay Transparency

Siemens follows Pay Transparency laws.

California Privacy Notice

California residents have the right to receive additional notices about their personal information. To learn more, [click here](#).

Criminal History

Qualified applications with arrest or conviction records will be considered for employment in accordance with applicable local and state laws.